

$$3.2. a) \|m_1\| = \frac{1}{\sqrt{\int_1^5 1 dx}} = \frac{1}{2}$$

$$v_2 = \langle m_1, v_2 \rangle m_1 = x - \left(\int_1^5 \frac{1}{2} x dx \right) \frac{1}{2} = x - 3$$

$$m_2 = \frac{x-3}{\|x-3\|} = \frac{x-3}{\frac{4}{\sqrt{3}}}$$

$$\|x-3\| = \sqrt{\int_1^5 (x-3)^2 dx} = \frac{4}{\sqrt{3}}$$

$$v_3 = \langle m_1, v_3 \rangle m_1 - \langle m_2, v_3 \rangle m_2 = x^2 - \left(\int_1^5 \frac{1}{2} x^2 dx \right) \frac{1}{2} - \left(\int_1^5 \frac{x-3}{\frac{4}{\sqrt{3}}} x^2 dx \right) \frac{x-3}{\frac{4}{\sqrt{3}}}$$

$$= x^2 - 6x + \frac{23}{3}$$

$$m_3 = \frac{\frac{3\sqrt{5}}{16} (x^2 - 6x + \frac{23}{3})}{\|x^2 - 6x + \frac{23}{3}\|}$$

$$\|x^2 - 6x + \frac{23}{3}\| = \sqrt{\int_1^5 (x^2 - 6x + \frac{23}{3})^2 dx} = \frac{3\sqrt{5}}{16}$$

$$\text{Basis } \left\{ \frac{1}{2}, \frac{x-3}{\frac{4}{\sqrt{3}}}, \frac{3\sqrt{5}}{16} (x^2 - 6x + \frac{23}{3}) \right\}$$

$$b) w = \int_1^5 \frac{1}{2} \sin dx \cdot \frac{1}{2} + \int_1^5 \frac{x-3}{\frac{4}{\sqrt{3}}} \sin dx \cdot \frac{x-3}{\frac{4}{\sqrt{3}}} + \int_1^5 \frac{3\sqrt{5}}{16} (x^2 - 6x + \frac{23}{3}) \sin dx \cdot \frac{3\sqrt{5}}{16} (x^2 - 6x + \frac{23}{3})$$

$$\text{da } w = \langle m_1, w \rangle m_1 + \langle m_2, w \rangle m_2 + \langle m_3, w \rangle m_3$$

$$= -0,0517x^2 - 0,3363x + 1,607$$

$$c) \| \sin x - (-0,0517x^2 - 0,3363x + 1,607) \| = \sqrt{\int_1^5 (\sin - (-0,0517x^2 - 0,3363x + 1,607))^2 dx} \\ = 0,443$$